

Dose-Based IV – Example 1

Order: Heparin 1,000 units/hr IV continuous

Available: Heparin 25,000 units in 500 mL NS (50 units/mL)

Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Step 2: Set up the dose rate as a fraction.

Step 3: Write the concentration as a conversion factor (units $\rightarrow$ mL).

Step 4: Multiply so that dose units cancel, leaving mL/hr.

Step 5: Calculate and program the pump.

Dose-Based IV – Example 1 Solution

Order: Heparin 1,000 units/hr IV continuous
Available: Heparin 25,000 units in 500 mL NS (50 units/mL)
Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Order: 1,000 units/hr Concentration: 50 units/mL

Step 2: Set up the dose rate as a fraction.

$$\frac{1,000 \text{ units}}{\text{hr}}$$

Step 3: Write the concentration as a conversion factor (units → mL).

$$\boxed{\frac{1 \text{ mL}}{50 \text{ units}}}$$

Step 4: Multiply so that dose units cancel, leaving mL/hr.

$$\frac{1,000 \cancel{\text{ units}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{50 \cancel{\text{ units}}}} = \text{——} \frac{\text{mL}}{\text{hr}}$$

Step 5: Calculate and program the pump.

$$\frac{1,000}{50} = 20 \frac{\text{mL}}{\text{hr}}$$

Set the IV pump to 20 mL/hr.

Dose-Based IV – Example 2

Order: Morphine sulfate 4 mg/hr IV continuous
Available: Morphine 10 mg in 5 mL NS (2 mg/mL)
Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Step 2: Set up the dose rate as a fraction.

Step 3: Write the concentration as a conversion factor (mg $\rightarrow$ mL).

Step 4: Multiply so that dose units cancel, leaving mL/hr.

Step 5: Calculate and program the pump.

Dose-Based IV – Example 2 Solution

Order: Morphine sulfate 4 mg/hr IV continuous
Available: Morphine 10 mg in 5 mL NS (2 mg/mL)
Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Order: 4 mg/hr Concentration: 2 mg/mL

Step 2: Set up the dose rate as a fraction.

$$\frac{4 \text{ mg}}{\text{hr}}$$

Step 3: Write the concentration as a conversion factor (mg → mL).

$$\boxed{\frac{1 \text{ mL}}{2 \text{ mg}}}$$

Step 4: Multiply so that dose units cancel, leaving mL/hr.

$$\frac{4 \cancel{\text{mg}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{2 \cancel{\text{mg}}}} = \text{ ______ } \frac{\text{mL}}{\text{hr}}$$

Step 5: Calculate and program the pump.

$$\frac{4}{2} = 2 \frac{\text{mL}}{\text{hr}}$$

Set the IV pump to 2 mL/hr.

Dose-Based IV – Example 3

Order: Potassium chloride 10 mEq/hr IV continuous
Available: KCl 40 mEq in 100 mL NS (0.4 mEq/mL)
Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Step 2: Set up the dose rate as a fraction.

Step 3: Write the concentration as a conversion factor (mEq $\rightarrow$ mL).

Step 4: Multiply so that dose units cancel, leaving mL/hr.

Step 5: Calculate and program the pump.

Dose-Based IV – Example 3 Solution

Order: Potassium chloride 10 mEq/hr IV continuous
Available: KCl 40 mEq in 100 mL NS (0.4 mEq/mL)
Question: What pump rate in mL/hr delivers the ordered dose?

Step 1: Identify the ordered dose rate and available concentration.

Order: 10 mEq/hr Concentration: 0.4 mEq/mL

Step 2: Set up the dose rate as a fraction.

$$\frac{10 \text{ mEq}}{\text{hr}}$$

Step 3: Write the concentration as a conversion factor (mEq → mL).

$$\boxed{\frac{1 \text{ mL}}{0.4 \text{ mEq}}}$$

Step 4: Multiply so that dose units cancel, leaving mL/hr.

$$\frac{10 \cancel{\text{ mEq}}}{\text{hr}} \times \boxed{\frac{1 \text{ mL}}{0.4 \cancel{\text{ mEq}}}} = \text{ ______ } \frac{\text{mL}}{\text{hr}}$$

Step 5: Calculate and program the pump.

$$\frac{10}{0.4} = 25 \frac{\text{mL}}{\text{hr}}$$

Set the IV pump to 25 mL/hr.